

Freshness Classification of Korean Food

Digital Image Processing

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Contents

1	Datasets	2
1.1	Available Datasets	2
1.2	Dataset Sources	2
1.3	Korean Food Dataset — Proposal	3
2	Methodology	3
2.1	Overall Pipeline	3
2.2	Backbone Architectures	4
2.3	Training Conditions	4
2.4	Label Modes	5
2.5	Evaluation Metrics	5
3	Preliminary Experiments	6
3.1	Setup	6
3.2	Results — Phase 1: ResNet-18 Baseline	6
3.3	Results — Phase 2: Backbone Comparison	6
3.4	All Runs — ResNet-18 Complete Results	7
3.5	Fruit-State Classification and Benchmark vs. Published Work	7
3.6	Model Ensemble	8
3.7	Confusion Matrix Analysis	9
3.8	Learning Curves	10
3.9	Domain Shift on Real Photos and Per-Fruit Analysis	10
4	Korean Food Dataset Proposal	11
4.1	Data Collection Pipeline	11
4.2	Integration with Training Pipeline	12
5	Software Implementation	12
5.1	Usage Workflow	12
5.2	<code>prepare_datasets.py</code> — Dataset Preparation	13
5.3	<code>backbone.py</code> — Backbone Registry	13
5.4	<code>dataset.py</code> — Data Loading	13
5.5	<code>train.py</code> — Training Loop	14
5.6	<code>analyze.py</code> — Visualization	14

1. Datasets

1.1. Available Datasets

Seven publicly available datasets were identified and surveyed. Table 1 summarizes their properties. Together they cover fruits, vegetables, and citrus varieties, with fresh/rotten binary labels and, in one case, a three-class annotation including formalin-mixed samples. One dataset (Ripening Stages) uses an object detection format (YOLOv5) rather than image classification and is therefore excluded from training experiments.

Table 1: Datasets surveyed. [†]Object detection format — excluded from classification experiments.

Dataset	Categories	Classes	Images	Size	Source
FruitVision	Apple, Banana, Grape, Mango, Orange	Fresh / Formalin / Rotten	10,154	857 MB	Mendeley
Lemon Varieties	7 lemon varieties	Fresh / Rotten	1,956	1.35 GB	Mendeley
Ripening Stages [†]	Strawberry, Avocado	Unripe / Partial / Ripe / Rotten	—	1.97 GB	Mendeley
Fruits Classification	Peach, Pomegranate, Strawberry	Fresh / Rotten	1,655	14.5 MB	Mendeley
Fruits Quality	12 mixed fruits	Fresh / Rotten	359	12.4 MB	Kaggle
Fruits Fresh/Rotten	Apple, Banana, Orange	Fresh / Rotten	13,599	1.95 GB	Kaggle
Fresh & Stale	Apple, Banana, Cucumber, Okra, Orange, Potato, Tomato	Fresh / Rotten	27,317	3.09 GB	Kaggle
Total (classification datasets)			55,040		

1.2. Dataset Sources

- **Fruit Vision** — <https://data.mendeley.com/datasets/xkbjx8959c/2>
- **Lemon Varieties** — <https://data.mendeley.com/datasets/mygrsk3vyb/2>
- **Ripening Stages** — <https://data.mendeley.com/datasets/mygrsk3vyb/2>
- **Fruits Classification** — <https://data.mendeley.com/datasets/rg254yr63x/1>
- **Fruits Quality** — <https://www.kaggle.com/datasets/nourabdoun/fruits-quality-fresh-v>

- **Fruits Fresh/Rotten** — <https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-ro>
- **Fresh & Stale** — <https://www.kaggle.com/datasets/swoyam2609/fresh-and-stale-classi>

1.3. Korean Food Dataset — Proposal

No labeled dataset for Korean food freshness currently exists in public repositories. Table 2 lists the target food categories and classification states for the proposed dataset.

Table 2: Target categories for the proposed Korean food freshness dataset.

Category	Romanization	States	Notes
Kimchi	kimchi	Fresh / Rotten	Color and mold during spoilage
Rice cake	tteok	Fresh / Rotten	Mold visible on surface
Side dishes	banchan	Fresh / Rotten	Vegetable-based, wilting
Cooked rice	bap	Fresh / Rotten	Mold indicator
Mandarin (Jeju)	gyul	Fresh / Semifresh / Rotten	Jeju variety, citrus
Strawberry	ttalgi	Fresh / Rotten	Fast spoilage

2. Methodology

2.1. Overall Pipeline

The classification pipeline consists of four sequential stages:

Stage	Component	Description
1	Data preparation	Raw datasets reorganized into unified food/state/ folder structure
2	Feature extraction	Pre-trained CNN backbone (ImageNet weights), frozen or partially unfrozen
3	Projection head	$d_{\text{out}} \rightarrow 256 \rightarrow 128$ linear projection with ReLU
4	Classification	Linear layer $128 \rightarrow C$ with cross-entropy loss

All input images are resized to 224×224 pixels. Training images undergo data augmentation (random resized crop, horizontal flip, $\pm 30^\circ$ rotation, color jitter on brightness/contrast/saturation/hue, Gaussian blur) to increase robustness to lighting and viewpoint variation. Validation images are center-cropped without augmentation. All images are normalized with ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

2.2. Backbone Architectures

Four backbone architectures are considered. Table 3 summarizes their properties. Experiments are conducted with ResNet-18, MobileNetV3-Small, EfficientNet-B0, and EfficientNet-B2. EfficientNet-B0 is identified as the recommended architecture for the final Korean food classifier based on its accuracy-to-parameter ratio and demonstrated suitability for small datasets.

Table 3: Backbone architectures considered in this study.

Model	Params	Out dim	Family
ResNet-18	11.2M	512	ResNet
MobileNetV3-S	2.5M	576	MobileNet
EfficientNet-B0	5.3M	1280	EfficientNet
EfficientNet-B2	9.1M	1408	EfficientNet

2.3. Training Conditions

Four training conditions form an ablation study to determine the optimal degree of backbone fine-tuning. Conditions C3 and C4 (with projection head) are the focus of this study, as the additional non-linear projection consistently improves performance over a direct linear probe.

Table 4: Training conditions (ablation study).

ID	Name	Backbone	Head	Description
C1	frozen	Frozen	Linear only	No backbone adaptation
C2	layer4	Layer4 free	Linear only	Partial fine-tuning, no projection
C3	head_frozen	Frozen	Projection + Linear	Full head, frozen backbone
C4	head_layer4	Layer4 free	Projection + Linear	Full head + partial fine-tuning

In all conditions the backbone is initialized with ImageNet pre-trained weights and produces a feature vector of dimension d_{out} (e.g. 512 for ResNet-18, Table 3).

C1 – frozen backbone, linear head. All backbone weights are frozen; gradients do not flow through the convolutional layers. A single linear layer $d_{\text{out}} \rightarrow C$ is trained from scratch. This is the classical *linear probe* setting, where the pre-trained backbone acts purely as a fixed feature extractor.

C2 – layer4 unfrozen, linear head. The last residual block of the backbone (layer4 in ResNet notation, comprising two residual units totaling $\approx 2\text{M}$ parameters for ResNet-18) is unfrozen and fine-tuned jointly with the linear head. All earlier layers remain frozen. This allows the backbone to specialize its high-level features to the target domain without the cost of full fine-tuning.

C3 – frozen backbone, projection head. A two-layer non-linear projection head ($d_{\text{out}} \rightarrow 256 \rightarrow 128$ with ReLU) is appended before the final linear classifier.

The backbone remains completely frozen; only the head is trained. The additional capacity of the projection layers allows the model to learn a more discriminative embedding space without modifying the backbone.

C4 – layer4 unfrozen, projection head. Combines the projection head of C3 with the partial fine-tuning of C2. The last backbone block (**layer4**) and the full projection head are optimized jointly, with a lower learning rate applied to the backbone ($\text{lr}_{\text{backbone}} = 10^{-5}$) to prevent catastrophic forgetting. This is the most expressive configuration and achieves the best validation F1 for ResNet-18. A key empirical finding of this study (Section 3.5) is that the optimal condition is backbone-dependent: ResNet-18 favors C4, whereas MobileNetV3 and both EfficientNet variants consistently peak at C3 (frozen backbone + projection head), where their richer feature vectors are already discriminative enough that backbone fine-tuning offers no further gain and only risks overfitting.

All models are trained with Adam optimizer ($\text{lr} = 10^{-3}$, $\text{lr}_{\text{backbone}} = 10^{-5}$, weight decay = 10^{-4}) and cosine annealing over 50 epochs. Batch size is 32.

2.4. Label Modes

Two classification targets are evaluated:

- **state**: classes correspond to freshness states only (fresh / rotten, or fresh / formalin / rotten). The model learns state-invariant features across all food categories.
- **fruit_state**: classes correspond to food-state combinations (e.g., **apple_fresh**, **apple_rotten**). The model simultaneously learns food identity and freshness state, enabling a richer confusion matrix analysis.

2.5. Evaluation Metrics

Given the potential class imbalance in real-world Korean food datasets, accuracy alone is insufficient. Let TP_c , FP_c , FN_c , and TN_c denote true positives, false positives, false negatives, and true negatives for class c , and let C be the total number of classes. The following metrics are reported:

- **Accuracy**: fraction of all samples correctly classified,

$$\text{Accuracy} = \frac{\sum_{c=1}^C TP_c}{\text{total samples}}. \quad (1)$$

- **Precision** (per class): fraction of predicted positives that are truly positive,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}. \quad (2)$$

- **Recall** (per class): fraction of actual positives correctly identified,

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}. \quad (3)$$

- **F1-score** (per class): harmonic mean of precision and recall,

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} = \frac{2TP_c}{2TP_c + FP_c + FN_c}. \quad (4)$$

- **Macro averages**: each per-class metric is averaged uniformly over all classes,

$$\overline{F1} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad \overline{\text{Precision}} = \frac{1}{C} \sum_{c=1}^C \text{Precision}_c, \quad \overline{\text{Recall}} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c. \quad (5)$$

Macro averaging treats every class equally regardless of sample count, which is essential when minority classes (e.g. formalin) are safety-critical.

- **Confusion matrix**: row-normalized per true class, to identify systematic misclassification patterns.

For food safety applications, recall on the *rotten* class is particularly critical: a false negative (rotten food classified as fresh) carries higher risk than a false positive.

3. Preliminary Experiments

3.1. Setup

Experiments are organized in three phases. **Phase 1** uses ResNet-18 as baseline backbone across all six datasets, all four conditions (C1–C4), and both label modes (state and fruit_state). **Phase 2** introduces MobileNetV3-Small, EfficientNet-B0, and EfficientNet-B2 for backbone comparison across conditions C1–C4. **Phase 3** targets the two apple/banana/orange and peach/pomegranate/strawberry collections (KFR and MFR) in fruit_state mode (6 classes each) with all four backbones, then combines them into ensembles and benchmarks the result against published work on the same datasets (Section 3.5). All experiments apply a stratified 80/20 train/validation split with fixed seed 42 and train for 60 epochs with Adam optimizer and cosine annealing.

3.2. Results — Phase 1: ResNet-18 Baseline

Table 5 reports the best macro F1-score for ResNet-18 across all datasets and conditions. C4 (head_layer4) achieves the best result in most datasets; C2 (layer4) wins on the smallest dataset (Fruits Quality, 359 images).

3.3. Results — Phase 2: Backbone Comparison

Table 6 compares the best F1-score per dataset across all four backbones (state mode, best condition per cell). The condition that produced each value is shown in parentheses.

Optimal condition per backbone. A consistent pattern emerges across datasets: ResNet-18 performs best under C4 (layer4 unfrozen + projection head) in most datasets, while MobileNetV3-Small and both EfficientNet variants perform best under C3 (backbone frozen + projection head). This suggests that the higher-dimensional feature vectors of the MobileNet/EfficientNet backbones (576–1408

Table 5: Best macro F1-score (%) — ResNet-18, all conditions, state mode. Bold: best per dataset.

Dataset	C1	C2	C3	C4	Best cond.
Fruits Fresh/Rotten	97.6	99.8	99.4	99.8	C2 / C4
Fresh & Stale	95.0	98.9	98.0	99.0	C4
Lemon Varieties	96.2	98.0	96.2	98.5	C4
Fruits Classification	94.3	95.5	95.2	96.4	C4
FruitVision (3-cls)	78.3	91.0	86.7	90.6	C2
Fruits Quality	86.1	91.6	86.0	88.9	C2

Table 6: Best macro F1-score (%) by backbone — state mode. Best condition shown in parentheses. Bold: best backbone per dataset.

Dataset	R18	MN3	EB0	EB2	Winner
Fruits Fresh/Rotten	99.8 (C4)	100.0 (C3)	100.0 (C3)	99.9 (C3)	MN3/EB0
Fresh & Stale	99.0 (C4)	99.2 (C3)	99.4 (C3)	99.3 (C3)	EB0
Lemon Varieties	98.5 (C4)	98.2 (C3)	98.7 (C3)	98.0 (C3)	EB0
Fruits Classification	96.4 (C4)	96.7 (C3)	97.0 (C3)	96.4 (C3)	EB0
FruitVision (3-cls)	91.0 (C2)	92.4 (C3)	91.6 (C3)	92.3 (C3)	MN3
Fruits Quality	91.6 (C2)	91.6 (C1)	86.1 (C4)	88.9 (C1)	R18/MN3

vs. 512 for ResNet-18) are already sufficiently discriminative without fine-tuning, whereas ResNet-18 benefits from partial backbone adaptation to the target domain.

Effect of dataset size. On the smallest dataset (Fruits Quality, 359 images), ResNet-18 outperforms EfficientNet-B0 by 5.5 percentage points. On datasets with more than 1,500 images, EfficientNet-B0 consistently matches or surpasses ResNet-18. This indicates that EfficientNet’s larger head is more prone to overfitting on very small datasets.

3.4. All Runs — ResNet-18 Complete Results

Table 7 lists all ResNet-18 runs sorted by F1-score.

3.5. Fruit-State Classification and Benchmark vs. Published Work

To position this work against the literature, the two apple/banana/orange (KFR) and peach/pomegranate/strawberry (MFR) collections were trained in `fruit_state` mode (6 classes: each fruit $\times$ fresh/rotten) with the full 4×4 grid of backbones and conditions. These two datasets were selected because published papers report results on the exact same data, enabling a like-for-like comparison. Tables 8 and 9 report the macro F1-score for every backbone–condition combination.

The C3 dominance noted for state mode holds even more strongly here: C3 is the best condition for every backbone except ResNet-18 on both datasets, and the two best individual models overall (EfficientNet-B0–C3 and MobileNetV3–C3) reach a

Table 7: All ResNet-18 runs sorted by F1-score (state mode, 60 epochs).

Dataset	Condition	Acc	F1	MCC	AUC
Fruits Fresh/Rotten	layer4	99.8	99.8	0.996	1.000
Fruits Fresh/Rotten	head_layer4	99.8	99.8	0.996	1.000
Fresh & Stale	head_layer4	99.0	99.0	0.979	1.000
Lemon Varieties	head_layer4	98.5	98.5	0.970	1.000
Fresh & Stale	layer4	98.9	98.9	0.978	1.000
Fruits Fresh/Rotten	head_frozen	99.4	99.4	0.987	0.999
Fresh & Stale	head_frozen	98.0	98.0	0.960	1.000
Lemon Varieties	layer4	98.0	98.0	0.960	0.999
Fruits Fresh/Rotten	frozen	97.6	97.6	0.952	0.997
Fresh & Stale	frozen	95.0	95.0	0.899	1.000
Fruits Classification	head_layer4	96.4	96.4	0.929	0.995
Lemon Varieties	frozen	96.2	96.2	0.925	0.992
Fruits Classification	layer4	95.5	95.5	0.912	0.993
Fruits Classification	head_frozen	95.2	95.2	0.904	0.990
Fruits Classification	frozen	94.3	94.3	0.888	0.982
FruitVision	layer4	90.9	91.0	0.863	0.981
FruitVision	head_layer4	90.4	90.6	0.856	0.982
Fruits Quality	layer4	91.7	91.6	0.839	0.947
FruitVision	head_frozen	86.4	86.7	0.797	0.965
Fruits Quality	head_layer4	88.9	88.9	0.789	0.951
Fruits Quality	frozen	86.1	86.1	0.732	0.914
FruitVision	frozen	78.0	78.3	0.686	0.932

Table 8: KFR (apple/banana/orange) — macro F1 (%), fruit_state mode, all backbones × conditions. Bold: best per backbone.

Backbone	C1	C2	C3	C4
ResNet-18	97.78	99.46	99.40	99.65
MobileNetV3-S	98.71	97.25	100.0	99.20
EfficientNet-B0	99.70	98.75	100.0	99.46
EfficientNet-B2	99.53	98.46	99.97	99.36

perfect 100.0% macro F1 on KFR. On the smaller, harder MFR collection the same EfficientNet-B0–C3 leads at 95.72%.

Comparison with published papers. All three reference papers report *accuracy* as their primary metric, so for a fair comparison our accuracy is used in Table 10 rather than macro F1. On both datasets our models surpass the published state of the art.

3.6. Model Ensemble

To test whether combining backbones improves on any single model, the 16 **fruit_state** models per dataset (4 architectures × 4 conditions) were combined by averaging their softmax outputs. Three weighting schemes were evaluated: **mean** (uniform average), **weighted** (by each model’s validation F1), and **adaptive** (per-image

Table 9: MFR (peach/pomegranate/strawberry) — macro F1 (%), fruit_state mode, all backbones × conditions. Bold: best per backbone.

Backbone	C1	C2	C3	C4
ResNet-18	90.98	92.67	94.09	93.35
MobileNetV3-S	93.13	89.48	95.25	93.84
EfficientNet-B0	93.36	84.05	95.72	88.49
EfficientNet-B2	92.82	82.08	93.98	84.71

Table 10: Benchmark against published work on the same datasets (metric: accuracy). Bold: best result per dataset.

Source / Model	Dataset / split	Acc.	Notes
<i>KFR — apple / banana / orange (6 classes)</i>			
Palakodati et al. (2020), CNN	5,989 / 60-10-30	97.82	partial KFR
Chakraborty et al. (2021), MobileNetV2	13,599 / 80-20	99.61	full KFR
Ours — best single (EB0/MN3-C3)	13,599 / 80-20	100.0	this work
Ours — ensemble (16 models)	13,599 / 80-20	99.96	this work
<i>MFR — peach / pomegranate / strawberry (6 classes)</i>			
Sharma & Kumar (2025), ResNet50	1,655 / 70-15-15	95.00	full MFR
Ours — best single (EB0-C3)	1,655 / 80-20	95.77	this work
Ours — ensemble (16 models)	1,655 / 80-20	96.68	this work

weighting by each model’s own confidence). The ensemble is computed purely from the per-image probability files written by the evaluation stage, requiring no GPU or model reloading. Table 11 reports the results.

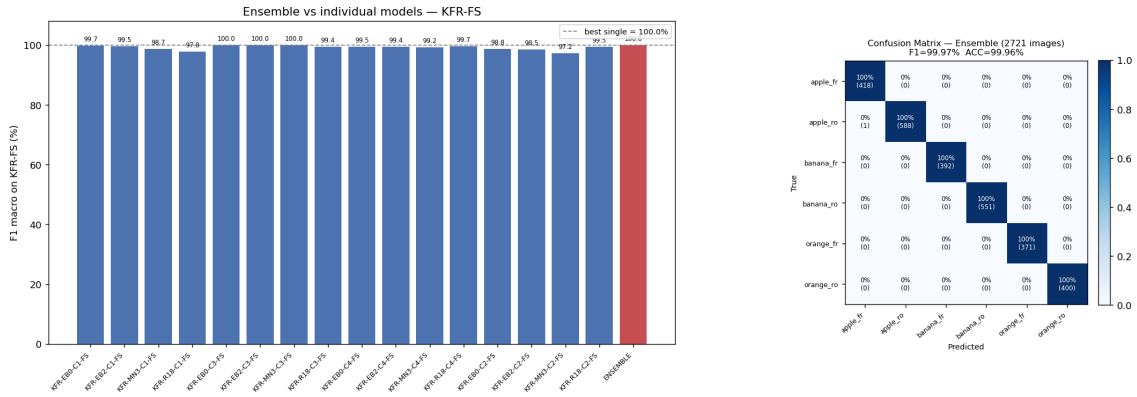
The result is informative in both directions and largely invariant to the weighting scheme. On KFR, where the best single model already reaches a perfect 100.0% F1, the ensemble cannot improve and in fact dips marginally (−0.03 pts): averaging a saturated model with weaker members can only pull the score down. On MFR, where substantial headroom remains, the ensemble adds +0.85 pts of F1 (+0.96 pts of accuracy over the best single model), confirming that the architectural diversity of four distinct backbone families yields genuinely complementary errors when the task is not already solved. Figures 1 and 2 show the per-model F1 bars alongside the ensemble confusion matrix for each dataset.

3.7. Confusion Matrix Analysis

Figure 3 shows the row-normalized confusion matrices for two representative completed experiments.

Table 11: Ensemble of 16 models per dataset (fruit_state mode). $\Delta F1$ is relative to the best single model.

Dataset	Method	Best single	Ens. F1	Ens. Acc	$\Delta F1$	MCC
KFR	mean	100.0	99.97	99.96	−0.03	0.9996
KFR	weighted	100.0	99.97	99.96	−0.03	0.9996
KFR	adaptive	100.0	99.97	99.96	−0.03	0.9996
MFR	mean	95.72	96.56	96.68	+0.84	0.9601
MFR	weighted	95.72	96.56	96.68	+0.84	0.9601
MFR	adaptive	95.72	96.57	96.68	+0.85	0.9601

**Figure 1:** KFR fruit_state ensemble (16 models, mean softmax). Left: macro F1 of each individual model and the ensemble. Right: row-normalized confusion matrix of the ensemble (percentages and counts), showing an essentially perfect diagonal.

3.8. Learning Curves

Figures 4–7 show the per-epoch metrics for all completed runs, grouped by dataset. Each curve corresponds to one training condition and backbone combination.

3.9. Domain Shift on Real Photos and Per-Fruit Analysis

To measure domain shift, trained models are evaluated on `own_dataset`, a collection of 219 real photos taken with a phone camera under natural lighting: 121 strawberry images (64 fresh, 57 rotten) and 98 banana images (46 fresh, 52 rotten). This set is used *only* for evaluation — never for training — and every photo is scored (no held-out split).

Per-fruit generalization tracks training exposure. Because `own_dataset` now contains two fruits, evaluation can be broken down per fruit (`03_05_per_fruit.py`). Across all 80 evaluated state-mode models a clear pattern emerges (Table 12): a model’s accuracy on a given fruit depends heavily on whether that fruit appeared in its training data. Models whose source dataset *included* banana (e.g. KFR, KFS, Fruits Quality) average 71.4% F1 on banana, whereas models that *never saw* banana (e.g. Lemon Varieties, peach/pomegranate/strawberry) collapse to 42.0% — barely above chance for the binary task. Strawberry F1 is essentially unaffected ($\approx 80\%$ either way), because the spoilage cues for strawberry transfer more readily.

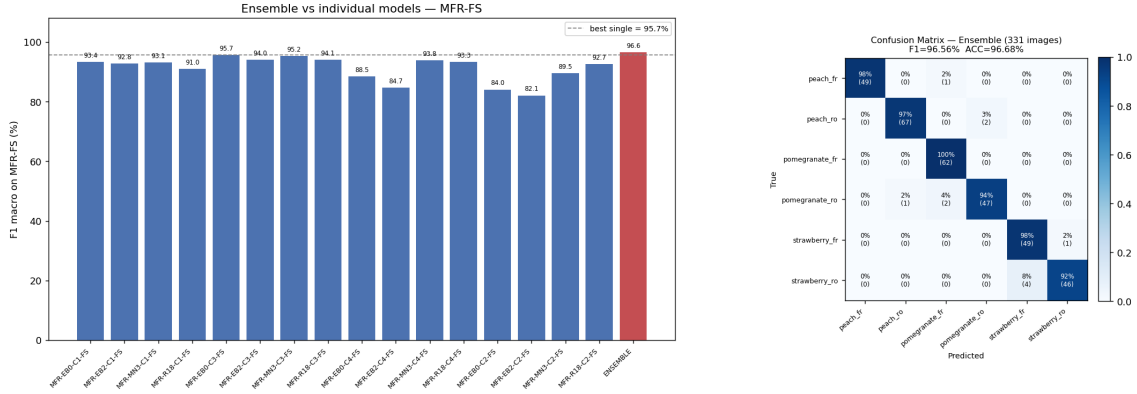


Figure 2: MFR fruit_state ensemble (16 models, mean softmax). The residual off-diagonal mass concentrates on the visually similar pomegranate/strawberry rotten states, consistent with the lower per-class F1 on those categories.

Table 12: Per-fruit F1 on own_dataset (219 photos, state mode), averaged over the 80 evaluated models, grouped by whether banana was seen during training.

Model group	# models	Banana F1	Strawberry F1
Saw banana in training	48	71.4	80.5
Did not see banana	32	42.0	79.9

The best all-round model is EfficientNet-B0–C3 trained on Fresh & Stale (banana 78%, strawberry 93%, 86.7% overall), reflecting that the broadest and largest training set transfers best to unseen photographic conditions. A per-image error analysis (03_04_error_analysis.py) over the same 219 photos and 80 models finds an accuracy ceiling of 100% with zero images misclassified by every model, confirming there are no inherently unlabelable photos — every image is correctly handled by at least one model.

Domain-adaptation fine-tuning. The domain gap can be closed cheaply. Taking the EfficientNet-B0 head trained on Fruits Quality (KFQ-EB0-C1-ST) as a starting point, the classifier head was fine-tuned on `own_dataset` with the backbone kept frozen (30 epochs, $\text{lr}_{\text{head}} = 5 \times 10^{-4}$, batch size 16), evaluated with 5-fold cross-validation. Macro F1 rises from a baseline of **86.3%** to **94.5%** (+8.2 pts), with the recall on the safety-critical rotten class improving in step. Because only the lightweight head is updated, this adaptation requires seconds of training and no backbone storage, making it a practical recipe for the eventual Korean food deployment: train on abundant public images, then fine-tune the head on a small set of in-domain photos.

4. Korean Food Dataset Proposal

4.1. Data Collection Pipeline

Two automated image collection scripts have been implemented, one targeting Korean fruits and one targeting Korean prepared foods. Both use Bing Image Search with 5–8 search queries per class per freshness state, combining Korean-language and

romanized search terms to maximize result diversity (e.g., “*fresh kimchi bright red color*” and “*moldy kimchi white fungus*” for the kimchi category).

Downloaded images are validated and processed as follows:

- **Integrity check:** corrupt or unreadable files are discarded.
- **Minimum resolution:** 100×100 pixels.
- **Format:** any format accepted; converted to RGB JPEG.
- **Output size:** 256×256 pixels (model crops to 224×224 during training).

4.2. Integration with Training Pipeline

The Korean food dataset will be organized in the same `food/state/` nested structure used throughout this study, making it directly compatible with the existing training and evaluation pipeline without any code modifications. The `prepare_datasets.py` script can be extended with a new preparation function, and the training pipeline selects the dataset interactively at runtime.

5. Software Implementation

The classification system is implemented in Python (PyTorch) and organized into four numbered stages (data preparation, training, evaluation/analysis, and reporting) plus two internal libraries. Figure 8 illustrates the data flow between components.

5.1. Usage Workflow

Scripts are organized in four numbered stages. Internal libraries (`_backbone.py`, `_dataset.py`) are imported automatically and never run directly.

1. **Stage 01 — Data preparation** `01_01_prepare_datasets.py` reorganizes raw datasets into the unified `data/{dataset}/{fruit}/{state}/` structure. `01_02_rename_own_dataset.py` and `01_03_prepare_own_dataset.py` handle own photos.
2. **Stage 02 — Training** `02_01_train.py` presents an interactive menu to select dataset, label mode, condition, and backbone. Results are saved to `run_outputs/{run_name}/`.
3. **Stage 03 — Evaluation and analysis** `03_01_evaluate.py` loads any `best_model.pt` and evaluates it on any dataset, generating per-image predictions and plots. `03_02_analyze.py` reads all `metrics.json` files and generates training-curve comparison plots. `03_03_ensemble.py` combines several models’ per-image probabilities into an ensemble (mean / weighted / adaptive) for any dataset and label mode, with a confusion-matrix plot. `03_04_error_analysis.py` ranks individual photos by how many models misclassify them, and `03_05_per_fruit.py` breaks the own_dataset evaluation down per fruit.
4. **Stage 04 — Reporting** `04_01_generate_tracker.py` produces `04_01_experiments_tracker` with color-coded run status. `04_02_generate_eval_report.py` produces

`04_02_eval_report.xlsx` with ranked cross-dataset results. `04_03_generate_own_report.py` consolidates the `own_dataset` domain-adaptation findings (fine-tuning, per-fruit, error analysis, ensemble) into `04_03_own_dataset_report.xlsx`, and `04_04_generate_benchmark_report.py` builds `04_04_benchmark_report.xlsx` comparing our models against the published papers of Section 3.5. Domain-adaptation fine-tuning itself is performed by `02_02_finetune.py`.

5.2. `prepare_datasets.py` — Dataset Preparation

Raw datasets are downloaded from their respective sources and placed in the `datasets/` directory, each in its original folder structure. `prepare_datasets.py` reads each raw dataset and reorganizes it into a unified `data/{dataset}/{fruit}/{state}/` hierarchy compatible with PyTorch’s `ImageFolder` loader. The script handles format differences across datasets: flat structures (all images in one folder), split structures (pre-divided `train/test/`), and nested structures (fruit then state sub-folders). Each dataset has a dedicated preparation function that normalizes class names, resolves naming inconsistencies (e.g. `freshpatato` $\rightarrow$ `potato/fresh`), and copies or hard-links images to the output directory. The script is idempotent: re-running it on an already-prepared dataset overwrites nothing.

5.3. `backbone.py` — Backbone Registry

A central registry (`BACKBONE_REGISTRY`) maps string identifiers ("`resnet18`", "`efficientnet_b0`", etc.) to factory functions that return pre-trained models loaded from `torchvision`. The `GenericBackbone` wrapper exposes a unified interface:

- `forward(x)` returns the feature vector of dimension d_{out} (the output of the backbone’s global pooling layer, *before* the original classification head).
- `unfreeze_layers(n)` selectively unfreezes the last n layer groups of the backbone, leaving all earlier weights frozen. For ResNet this corresponds to the last n residual stages (`layer4`, `layer3`, ...).

This design decouples the backbone choice from the training loop, making it trivial to swap architectures.

5.4. `dataset.py` — Data Loading

`dataset.py` provides three loaders that cover the three folder layouts found across the six datasets:

- **Layout A** (flat): all images in a single directory.
- **Layout B** (split): pre-divided `train/` and `test/` sub-folders.
- **Layout C** (nested): `fruit/state/` hierarchy, used for all experiments in this study.

For Layout C, `get_loaders_nested()` constructs a stratified 80/20 train/validation split using scikit-learn’s `StratifiedShuffleSplit`. Two separate `ImageFolder` objects (one with augmentation transforms, one with validation-only transforms) are created over the same index sets so that training and validation images receive different preprocessing. The `label_mode` parameter controls whether class labels

are derived from freshness state only (`state`) or from fruit-state combinations (`fruit_state`).

5.5. `train.py` — Training Loop

The training script presents an interactive menu for dataset, label mode, condition, and backbone selection. The run directory name is deterministically computed from these choices (e.g. `mendeley_fruitvision_all_head_layer4_resnet18_fs`), which serves as both a unique identifier and a resume key.

Training proceeds for a fixed number of epochs (default 50) with the following behavior:

- **Evaluation every 5 epochs:** accuracy, macro F1, macro precision, macro recall, per-class metrics, and confusion matrix are computed on the validation set and appended to `metrics.json`.
- **Rolling checkpoint every 10 epochs:** a single `checkpoint.pt` file is overwritten, keeping disk usage constant.
- **Best model:** whenever validation F1 improves, the full model state is saved to `best_model.pt`.
- **Resume:** if a checkpoint exists in the run directory, training automatically resumes from the last saved epoch.

At the end of training, all metrics are serialized to `metrics.json`, which serves as the single source of truth for all subsequent analysis.

5.6. `analyze.py` — Visualization

`analyze.py` scans `run_outputs/` for all `metrics.json` files, loads them into a list of run records, and generates nine plots grouped into three categories.

Learning curves (one subplot per dataset, one line per condition+backbone combination):

- `acc_curves.png` — validation accuracy vs. epoch. Shows whether training has converged and whether different conditions converge at different rates.
- `f1_curves.png` — macro F1 vs. epoch. The primary metric used for model selection; reveals overfitting when F1 peaks and then declines.
- `precision_curves.png` — macro precision vs. epoch. Indicates how often the model's positive predictions are correct.
- `recall_curves.png` — macro recall vs. epoch. Indicates how many true positives the model captures; particularly important for the rotten and formalin classes.
- `loss_curves.png` — training loss and validation loss vs. epoch on the same axes. Divergence between the two signals overfitting.

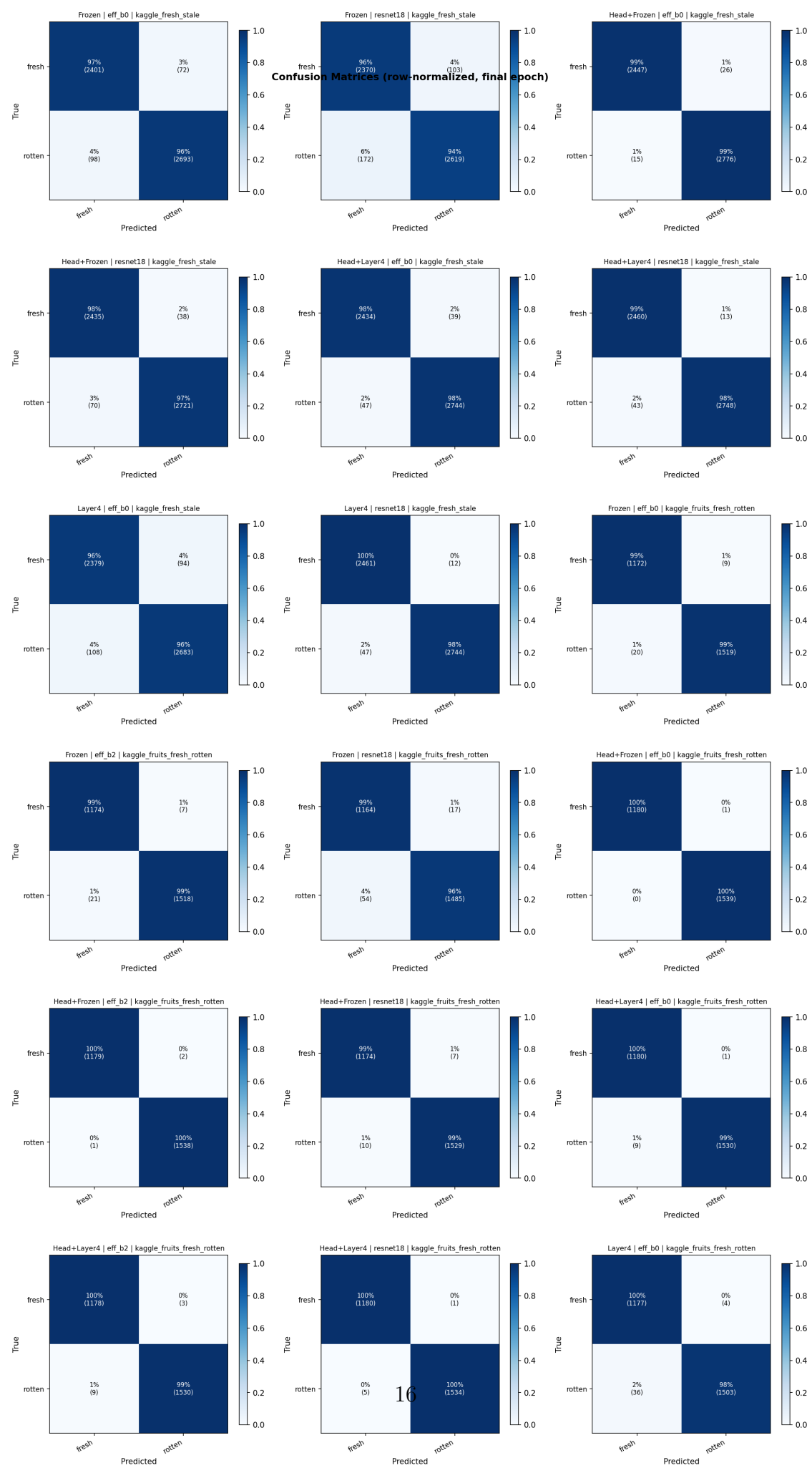
Aggregate comparisons (summarize best results across all runs):

- `metrics_summary.png` — grouped bar chart showing the best macro F1, precision, recall, and accuracy for each run, organized by dataset and condition. Useful for a quick overview of which configuration wins.
- `heatmap.png` — a condition $\times$ backbone grid where each cell shows the best F1 achieved. Empty cells correspond to experiments not yet run.

Error analysis (inspect what the model gets wrong):

- `confusion_matrix.png` — row-normalized confusion matrix for each completed run at the final epoch. Each row represents a true class; each column a predicted class. Values on the diagonal are correct classifications; off-diagonal values reveal systematic confusions (e.g. formalin predicted as fresh).
- `per_class.png` — bar chart of F1, precision, and recall broken down by individual class for each run. Identifies which specific class is pulling the macro average down.

All plots are saved to `run_outputs/_plots/`. Running `prepare_figures.py` copies them — together with the ensemble confusion matrices from `run_outputs/_ensemble/` and the per-fruit breakdown — into `figures/` for inclusion in this report.



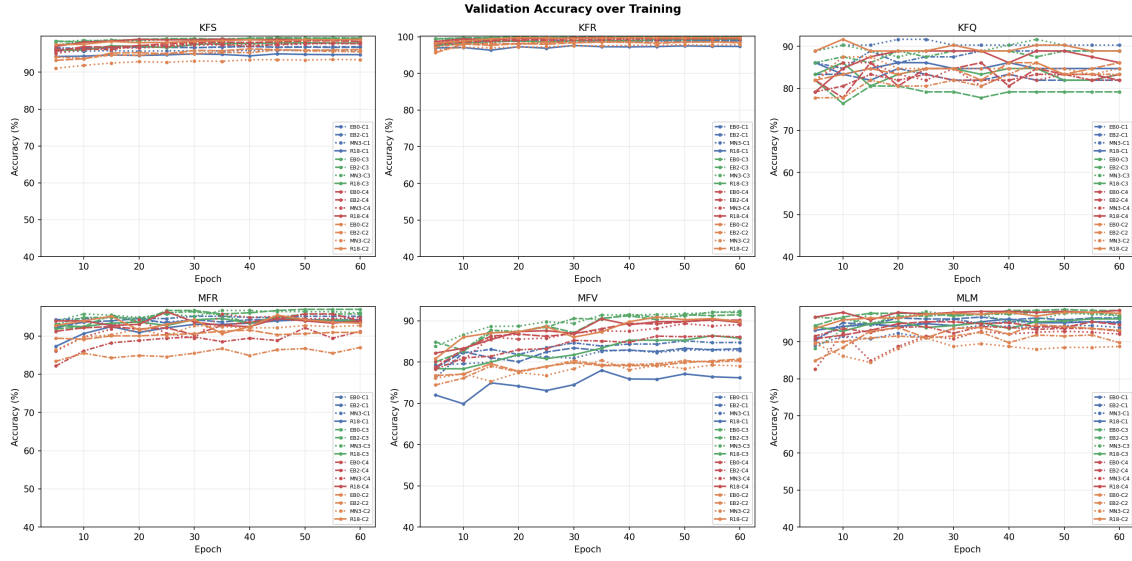


Figure 4: Validation accuracy over training epochs per dataset.

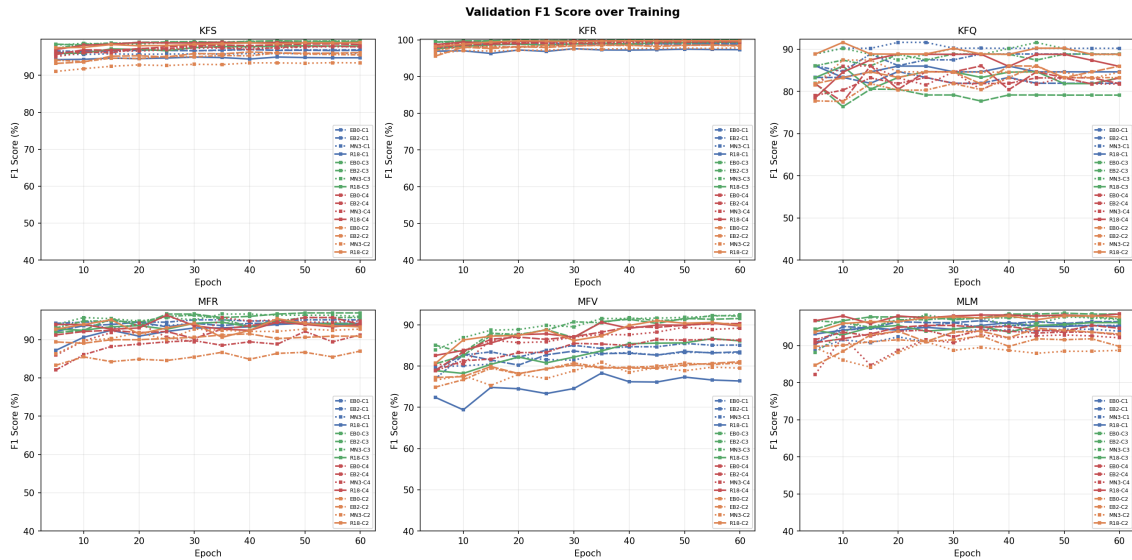


Figure 5: Macro-averaged F1 score over training epochs per dataset.

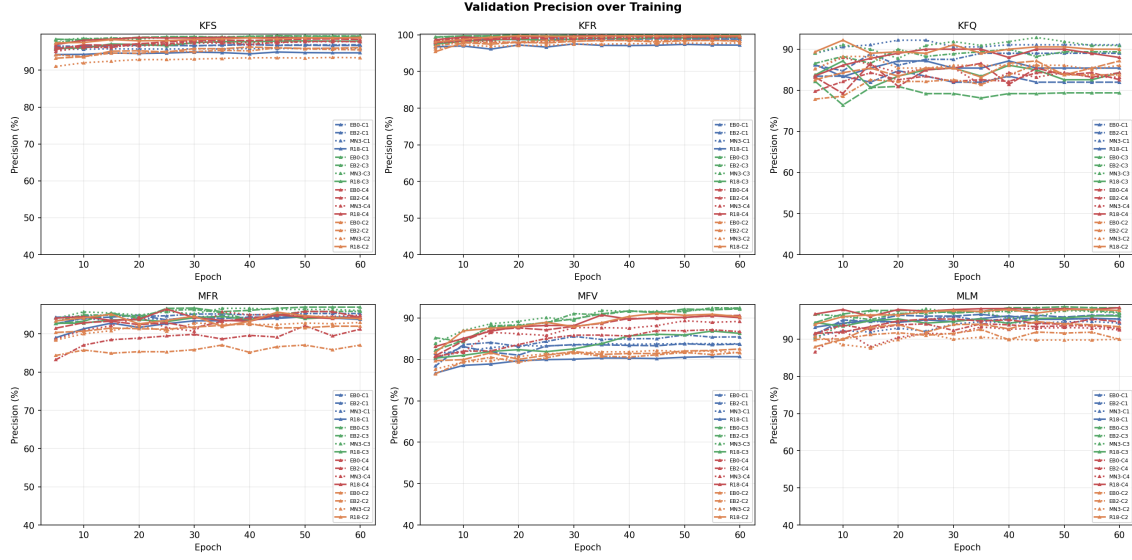


Figure 6: Macro-averaged precision over training epochs per dataset.

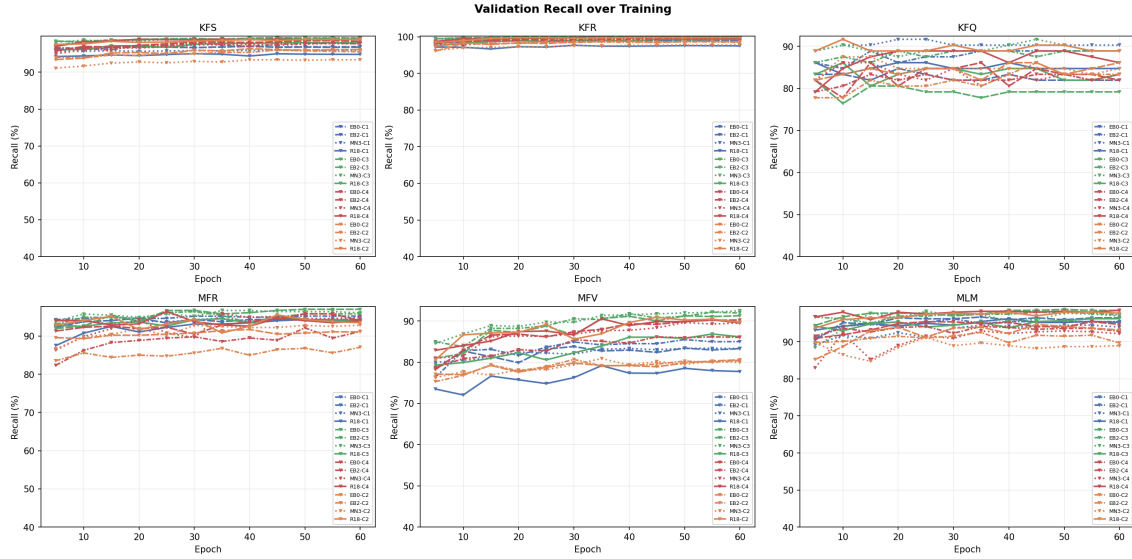


Figure 7: Macro-averaged recall over training epochs per dataset.

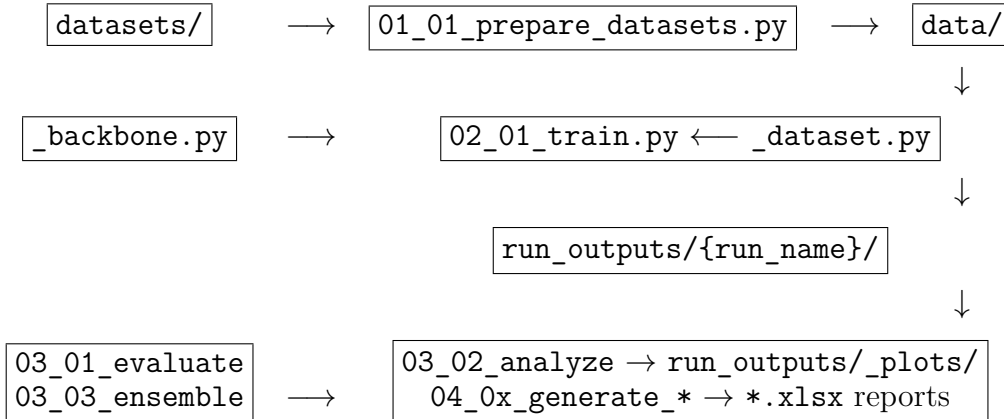


Figure 8: Data flow across the four numbered stages. Internal libraries (`_backbone.py`, `_dataset.py`) are imported, never run directly.